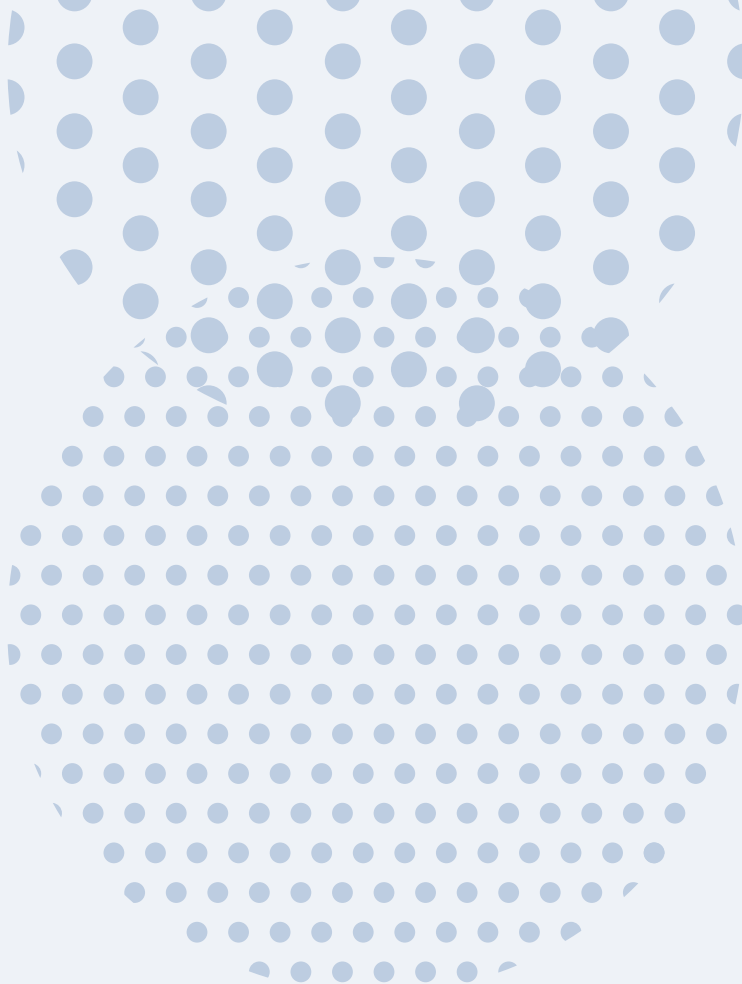




38.5 • A photon has momentum of magnitude $8.24 \times 10^{-28} \text{ kg} \cdot \text{m/s}$.
(a) What is the energy of this photon? Give your answer in joules and in electron volts. (b) What is the wavelength of this photon? In what region of the electromagnetic spectrum does it lie?

38.6 ●● A clean nickel surface is exposed to light of wavelength 206 nm. What is the maximum speed of the photoelectrons emitted from this surface? Use Table 38.1.

38.7 ●● When ultraviolet light with a wavelength of 400.0 nm falls on a certain metal surface, the maximum kinetic energy of the emitted photoelectrons is measured to be 1.10 eV. What is the maximum kinetic energy of the photoelectrons when light of wavelength 300.0 nm falls on the same surface?

38.8 ●● The photoelectric work function of potassium is 2.3 eV. If light that has a wavelength of 260 nm falls on potassium, find (a) the stopping potential in volts; (b) the kinetic energy, in electron volts, of the most energetic electrons ejected; (c) the speed of these electrons.

38.9 • When ultraviolet light with a wavelength of 254 nm falls on a clean copper surface, the stopping potential necessary to stop emission of photoelectrons is 0.181 V. (a) What is the photoelectric threshold wavelength for this copper surface? (b) What is the work function for this surface, and how does your calculated value compare with that given in Table 38.1?

38.15 ●● X rays with initial wavelength 0.0665 nm undergo Compton scattering. What is the longest wavelength found in the scattered x rays? At which scattering angle is this wavelength observed?

38.16 ●● A photon with wavelength $\lambda = 0.1385$ nm scatters from an electron that is initially at rest. What must be the angle between the direction of propagation of the incident and scattered photons if the speed of the electron immediately after the collision is 8.90×10^6 m/s?

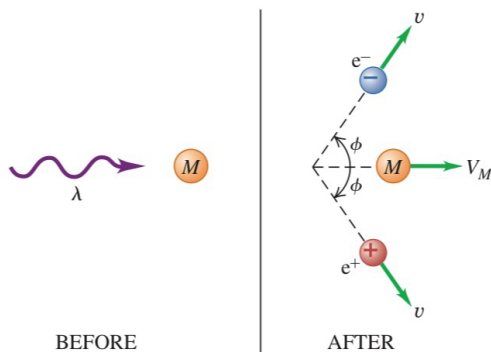
38.17 •• If a photon of wavelength 0.04250 nm strikes a free electron and is scattered at an angle of 35.0° from its original direction, find (a) the change in the wavelength of this photon; (b) the wavelength of the scattered light; (c) the change in energy of the photon (is it a loss or a gain?); (d) the energy gained by the electron.

38.28 •• CP A photon with wavelength $\lambda = 0.0780 \text{ nm}$ is incident on an electron that is initially at rest. If the photon scatters in the backward direction, what is the magnitude of the linear momentum of the electron just after the collision with the photon?

38.33 ●● A photon with wavelength 0.1120 nm collides with a free electron that is initially at rest. After the collision the wavelength is 0.1140 nm. (a) What is the kinetic energy of the electron after the collision? What is its speed? (b) If the electron is suddenly stopped (for example, in a solid target), all of its kinetic energy is used to create a photon. What is the wavelength of this photon?

38.39 ••• CP A photon with wavelength λ is incident on a stationary particle with mass M , as shown in **Fig. P38.39**. The photon is annihilated while an electron–positron pair is produced. The target particle moves off in the original direction of the photon with speed V_M . The electron travels with speed v at angle ϕ with respect to that direction. Owing to momentum conservation, the positron has the same speed as the electron. (a) Using the notation $\gamma = (1 - v^2/c^2)^{-1/2}$ and $\gamma_M = (1 - V_M^2/c^2)^{-1/2}$, write a relativistic expression for energy conservation in this process. Use the symbol m for the electron mass. (b) Write an analogous expression for momentum conservation.

Figure P38.39



(c) Eliminate the ratio h/λ between your energy and momentum equations to derive a relationship between v/c , V_M/c , and ϕ in terms of the ratio m/M , keeping γ and γ_M as a useful shorthand notation. (d) Consider the case where $V_M \ll c$, and use a binomial expansion to derive an expression for γ_M to the first order in V_M . Use that result to rewrite your previous result as an expression for V_M in terms of c , v , m/M , and ϕ . (e) Are there choices of v and ϕ for which $V_M = 0$? (f) Suppose the target particle is a proton. If the electron and positron remain stationary, so that $v = 0$, then with what speed does the proton move, in km/s? (g) If the electron and positron each have total energy 5.00 MeV and move with $\phi = 60^\circ$, then what is the speed of the proton? (*Hint:* First solve for γ and v .) (h) What is the energy of the incident photon in this case?

38.40 ●●● Consider Compton scattering of a photon by a *moving* electron. Before the collision the photon has wavelength λ and is moving in the $+x$ -direction, and the electron is moving in the $-x$ -direction with total energy E (including its rest energy mc^2). The photon and electron collide head-on. After the collision, both are moving in the $-x$ -direction (that is, the photon has been scattered by 180°). (a) Derive an expression for the wavelength λ' of the scattered photon. Show that if $E \gg mc^2$, where m is the rest mass of the electron, your result reduces to

$$\lambda' = \frac{hc}{E} \left(1 + \frac{m^2 c^4 \lambda}{4hcE} \right)$$

(b) A beam of infrared radiation from a CO_2 laser ($\lambda = 10.6 \mu\text{m}$) collides head-on with a beam of electrons, each of total energy $E = 10.0 \text{ GeV}$ ($1 \text{ GeV} = 10^9 \text{ eV}$). Calculate the wavelength λ' of the scattered photons, assuming a 180° scattering angle. (c) What kind of scattered photons are these (infrared, microwave, ultraviolet, etc.)? Can you think of an application of this effect?

7. (a) What is the surface temperature of Betelgeuse, a red giant star in the constellation Orion (Fig. 40.4), which radiates with a peak wavelength of about 970 nm? (b) Rigel, a bluish-white star in Orion, radiates with a peak wavelength of 145 nm. Find the temperature of Rigel's surface.

8. An FM radio transmitter has a power output of 150 kW and operates at a frequency of 99.7 MHz. How many photons per second does the transmitter emit?

M

14. Show that at long wavelengths, Planck's radiation law (Eq. 40.6) reduces to the Rayleigh–Jeans law (Eq. 40.3).

15. A simple pendulum has a length of 1.00 m and a mass of 1.00 kg. The maximum horizontal displacement of the pendulum bob from equilibrium is 3.00 cm. Calculate the quantum number n for the pendulum.

- 19.** Two light sources are used in a photoelectric experiment to determine the work function for a particular metal surface. When green light from a mercury lamp ($\lambda = 546.1 \text{ nm}$) is used, a stopping potential of 0.376 V reduces the photocurrent to zero. (a) Based on this measurement, what is the work function for this metal? (b) What stopping potential would be observed when using the yellow light from a helium discharge tube ($\lambda = 587.5 \text{ nm}$)?

20. Lithium, beryllium, and mercury have work functions **W** of 2.30 eV, 3.90 eV, and 4.50 eV, respectively. Light with a wavelength of 400 nm is incident on each of these metals. (a) Determine which of these metals exhibit the photoelectric effect for this incident light. Explain your reasoning. (b) Find the maximum kinetic energy for the photoelectrons in each case.

- 29.** A 0.001 60-nm photon scatters from a free electron.
- M** For what (photon) scattering angle does the recoiling electron have kinetic energy equal to the energy of the scattered photon?

- 30.** After a 0.800-nm x-ray photon scatters from a free electron, the electron recoils at 1.40×10^6 m/s. (a) What is the Compton shift in the photon's wavelength? (b) Through what angle is the photon scattered?

- 32.** A photon having energy E_0 is scattered by a free electron initially at rest such that the scattering angle of the scattered electron is equal to that of the scattered photon as shown in Figure P40.31. (a) Determine the angle θ . (b) Determine the energy and momentum of the scattered photon. (c) Determine the kinetic energy and momentum of the scattered electron.

- 36.** Find the maximum fractional energy loss for a 0.511-MeV gamma ray that is Compton scattered from (a) a free electron and (b) a free proton.

43. In the Davisson–Germer experiment, 54.0-eV electrons were diffracted from a nickel lattice. If the first maximum in the diffraction pattern was observed at $\phi = 50.0^\circ$ (Fig. P40.43), what was the lattice spacing a between the vertical columns of atoms in the figure?

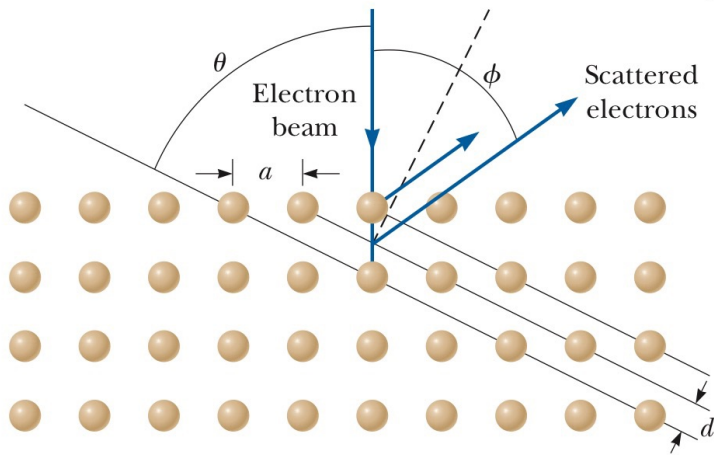


Figure P40.43

44. The nucleus of an atom is on the order of 10^{-14} m in diameter. For an electron to be confined to a nucleus, its de Broglie wavelength would have to be on this order of magnitude or smaller. (a) What would be the kinetic energy of an electron confined to this region? (b) Make an order-of-magnitude estimate of the electric potential energy of a system of an electron inside an atomic nucleus. (c) Would you expect to find an electron in a nucleus? Explain.

52. In a certain vacuum tube, electrons evaporate from a hot cathode at a slow, steady rate and accelerate from rest through a potential difference of 45.0 V. Then they travel 28.0 cm as they pass through an array of slits and fall on a screen to produce an interference pattern. If the beam current is below a certain value, only one electron at a time will be in flight in the tube. In this situation, the interference pattern still appears, showing that each individual electron can interfere with itself. What is the maximum value for the beam current that will result in only one electron at a time in flight in the tube?

- 55.** An electron and a 0.020 0-kg bullet each have a velocity of magnitude 500 m/s, accurate to within 0.010 0%. Within what lower limit could we determine the position of each object along the direction of the velocity?
- M**